CONFIGURE IPV6 ADDRESSING

TASK-1 Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Assign a Static IPv6 Address

Operating System: Windows

Commands used:

Step-1 ipconfig — to identify the network adapter and inspect its current configuration.

Step-2 netsh interface ipv6 add address interface="Wi-Fi" address=fd12:3456:789a:1::10/64 store=persistent — to assign a static IPv6 address.

Step-3 ipconfig — to verify the assigned IPv6 address.

Assigned IPv6 Address: fd12:3456:789a:1::10/64

Result: The static IPv6 address was configured on the selected network adapter and verified using the command line.

TASK-2 Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Hint: Use the ping6 or ping -6 command for IPv6 testing.

STEP :1 Check your network adapter – ipconfig

Step 2: Assign a Unique Local IPv6 address- netsh interface ipv6 add address interface="Wi-Fi" address=fd12:3456:789a:1::10/64

Step 3: Configure the second device - 1. fd12:3456:789a:1::10/64
2. fd12:3456:789a:1::20/64

Step 4: Verify and test connectivity – ipconfig - fd12:3456:789a:1::10

second device: ping -6 fd12:3456:789a:1::20

TASK-3 Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect .Constraint: Document the steps and the observed address behavior in a text file.

STEP:

1. Open Command Prompt.
2. Run ipconfig to inspect the current IPv6 configuration.
3. Enable temporary IPv6 address generation using:
   netsh interface ipv6 set privacy state=enabled
4. Check the setting using:
   netsh interface ipv6 show privacy
5. Record the temporary IPv6 address using ipconfig.
6. Disconnect and reconnect the network.
7. Run ipconfig again and compare the address.
8. Restart the computer, check again, and record the result.

Observations:

Initial temporary IPv6 address: 2409:40c1:110c:3cb1:fe13:2605:8f6c:845d

Address after network reconnection:

TASK-4 Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

CODE:

```
restaurants = [
   {"name": "Domino's Pizza", "ipv6": "fd12:3456:789a::1"},
   {"name": "Pizza Hut", "ipv6": "fd12:3456:789a::2"},
   {"name": "McDonald's", "ipv6": "fd12:3456:789a::3"},
   {"name": "Burger King", "ipv6": "fd12:3456:789a::4"},
   {"name": "Subway", "ipv6": "fd12:3456:789a::5"}
]

search = input("Enter restaurant name: ").lower()

found = False

for restaurant in restaurants:
   if search in restaurant["name"].lower():
      print("Restaurant:", restaurant["name"])
      print("IPv6 Address:", restaurant["ipv6"])
      found = True

if not found:
   print("No matching restaurant found.")
```

OUTPUT:

```
Enter restaurant name: pizza
```

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Restaurant: Domino's Pizza

IPv6 Address: fd12:3456:789a::1

Restaurant: Pizza Hut

IPv6 Address: fd12:3456:789a::2

TASK-5 Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

AI-Generated Scenario:
The college network uses the Unique Local Address prefix fd12:3456:789a::/48. It is divided into three /64 subnets:

1. Computer Lab: fd12:3456:789a:1::/64
2. Administration: fd12:3456:789a:2::/64
3. Library: fd12:3456:789a:3::/64

Configuration Steps:

1. Open Command Prompt as Administrator.
2. Identify the network interface using:
   netsh interface ipv6 show interfaces
3. Assign an IPv6 address from Subnet 1:
   netsh interface ipv6 add address interface="Wi-Fi" address=fd12:3456:789a:1::10/64
4. Verify the configuration using:
   ipconfig
5. Check that the assigned IPv6 address appears under the selected network adapter.